

Satellite-to-Ground Continuous Variable Quantum Key Distribution: The Gaussian and Discrete Modulated Protocols in Low Earth Orbit

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Abstract—The Gaussian modulated continuous variable quantum key distribution (GM-CVQKD) protocol is known to maximise the mutual information between two parties during quantum key distribution (QKD). An alternative modulation scheme is the discrete modulated CVQKD (DM-CVQKD) protocol. In this paper, we perform a feasibility study for practical satellite-to-ground CVQKD with a focus on comparing existing GM-CVQKD and DM-CVQKD protocols with the application of the performance of recent reconciliation methods. We use a satellite-to-ground link model which takes into account geometric, scintillation, and scattering losses from the link distance, atmospheric turbulence, and atmospheric aerosols, respectively. In addition, recent multidimensional (MD) and multilevel coding and multistage decoding (MLC-MSD) reconciliation method models in combination with multiedge-type low-density parity-check (MET-LDPC) code models have been used to determine the reconciliation efficiency. Realistic parameters have been used for the calculation of the secret key rate (SKR) along with a comparison between MD and MLC-MSD reconciliation during satellite-to-ground GM-CVQKD, for the first time. The results show that GM-CVQKD outperforms DM-CVQKD. In addition, GM-CVQKD with MD reconciliation outperforms GM-CVQKD with MLC-MSD reconciliation in the finite size limit by producing positive SKRs at larger link distances and lower elevation angles.

Index Terms—Quantum key distribution, continuous variable, Gaussian modulation, discrete modulation, satellite communication, quantum communication.

I. INTRODUCTION

QUANTUM key distribution (QKD) [1] is a method of sharing a secret key between two parties, Alice and Bob,

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where eavesdropping by Eve can be inferred as a consequence of fundamental quantum mechanics. The most significant advancement of QKD in space-based applications was the demonstration of discrete variable QKD (DVQKD), which uses the available degrees of freedom of single photons, to encode a key between optical ground stations on Earth and the Micius satellite [2]. Despite its successful deployment in space, its use of expensive and inefficient single photon detectors for detection poses challenges for its popularisation and commercialisation [3]. An appealing alternative is continuous variable QKD (CVQKD) which uses multi-photon technologies to encode the key in the continuous X and P quadratures of light [3], [4]. Its use of homodyne or heterodyne detection is more cost-effective, more compatible with standard telecommunication optical networks, and more efficient, offering higher secret key rates (SKRs) [3], [5], [6].

CVQKD experiments have predominantly been restricted to fibre-based systems in the laboratory where a SKR of 14.2 Mbit/s over 15 km of optical fibre has been demonstrated using a local oscillator [7]. The first demonstration of fibre-based CVQKD over 100 km was performed by controlling and suppressing excess noise [8]. A SKR of 2.1 Gbit/s in free-space has been achieved using 10-channel wavelength division multiplexing in the weak turbulence regime [9] using the Kramers-Kronig scheme which digitally separates the quadrature components [10], and a SKR greater than 1.68 Gbit/s can be reached in a lossless and excess noise free system that uses two polarisations, six wavelengths, and four orbital angular momentum for multiplexing [11]. In addition, an average SKR of 1.015 Mbit/s was achieved over a turbulent 1 m free-space channel with a transmittance between -6 dB to -15 dB using the passive-state-preparation CVQKD protocol, which has the same security proof as GM-CVQKD assuming Alice's transmitter is trusted [12]. A study of atmospheric effects on quantum communications over 1.6 km of free-space was performed and the experimental setup was capable of CVQKD [13]. However, there has only been one demonstration of free-space CVQKD which occurred over 460 m in an urban environment and achieved a SKR of 0.152 kbit/s using unidimensional Stokes operator encoding by modulating the quantum Stokes parameter with a Gaussian distribution [14]. It should be noted that the Kramers-Kronig scheme and Stokes operator encoding do not provide unconditional security, unlike GM-CVQKD [3].

The two modulation approaches for CVQKD to encode a key are Gaussian modulation (GM-CVQKD) [3] and

discrete modulation (DM-CVQKD) [15], where implementations can be found in associated publications, [16], [17]. The GM-CVQKD protocol is capable of achieving higher SKRs as the coherent state distribution follows a continuous Gaussian distribution allowing for more positions on the optical phase space [3]. In contrast, DM-CVQKD positions a finite number of coherent states on the optical phase space, limiting its achievable SKR [18], [19], [20]. Under ideal circumstances of perfect CVQKD and channel parameters, GM-CVQKD produces larger SKRs. However, it is not evident that DM-CVQKD performs better under realistic non-ideal parameters. In fact, DM-CVQKD was proposed to address the poor performance of CVQKD in the low SNR regimes [15]. Since the coherent states on the optical phase space follow a discretised distribution, it may be argued that the states are more distinguishable and can accommodate more excess noise.

Previous studies on satellite-to-ground CVQKD provide a variety of models for the laser propagation between a satellite and an optical ground station (OGS) [21], [22], [23], [24], [25]. However, there are no published comparisons between existing GM-CVQKD and DM-CVQKD protocols with the application of the performance of existing reconciliation methods for CVQKD in a satellite-to-ground context. In this work, the feasibility of practical low Earth orbit (LEO - orbit altitudes of 160-1000 km) satellite-to-ground GM-CVQKD and DM-CVQKD are investigated. These protocols are also compared in the asymptotic and finite size limits, using realistic parameters to calculate accessible SKRs. The DM-CVQKD protocols studied are the M -PSK protocol that assume Gaussian optimality under collective attacks [15], [18], [19], [20], [26], [27], and the M -QAM protocol which does not assume Gaussian optimality under collective attacks [28]. The calculated achievable asymptotic limit SKRs are compared to the known SKR upper bound in a satellite-to-ground context [29], [30]. In addition, the performance of recent multidimensional (MD) reconciliation and multilevel coding and multistage decoding (MLC-MSD) reconciliation methods are applied to the GM-CVQKD protocol for the calculation of finite size limit SKRs in a satellite-to-ground context.

The work is structured as follows. Section II introduces the GM-CVQKD protocol as well as the M -PSK and M -QAM DM-CVQKD protocol. Section III describes the process used when considering finite size effects. Section IV introduces the LEO satellite to OGS link model. Section V discusses the achievable SKRs of LEO satellite-to-ground GM-CVQKD and DM-CVQKD, and the resulting trends in SKR from parameter variation. Section VI discusses aspects of the feasibility study for real world satellite-to-ground CVQKD, and Section VII presents the conclusions.

II. THE GM-CVQKD AND DM-CVQKD PROTOCOLS

In this section, the GM-CVQKD protocol and the M -PSK DM-CVQKD protocol with security under collective attacks and assuming Gaussian optimality are investigated. The M -QAM DM-CVQKD protocol with security under collective attacks without assuming Gaussian optimality is also studied.

TABLE I

EXCESS NOISE IN DAYLIGHT [31]. THE VALUES ARE NORMALISED TO SHOT NOISE AND ARE EXPRESSED IN SHOT NOISE UNITS (SNU)

ϵ	Source	Value (SNU)
Channel Excess Noise, ϵ_{ch}	Time-of-arrival fluctuations	0.0060
	Atmospheric relative intensity noise in local oscillator	0.0100
	Relative intensity noise in local oscillator	0.0018
	Modulation noise	0.0005
	Background noise	0.0002
Detection Excess Noise, ϵ_{det}	Relative intensity noise in signal	0.0001
	Electronic noise	0.0130
	Anaogue-to-digital converter noise	0.0002
	Detector overlap	0.0001
	Local oscillator subtraction noise	0.0001
	Local oscillator to signal leakage	0.0001

A. GM-CVQKD

In GM-CVQKD [3], Alice sends a series of displaced coherent states,

$$|\alpha\rangle = |q + ip\rangle, \quad (1)$$

where q and p are the amplitude and phase quadrature components. These are both random variables of the same zero-centred normal distribution:

$$q, p = \mathcal{N}(0, V_A). \quad (2)$$

Here, V_A is the modulation variance. The covariance matrix describing the Gaussian modulated coherent state sent from Alice to Bob is

$$\gamma_{AB} = \begin{bmatrix} (V_A + 1)\mathbf{I} & \sqrt{T(V_A^2 + 2V_A)}\sigma_z \\ \sqrt{T(V_A^2 + 2V_A)}\sigma_z & T(V_A + 1 + \chi_{line})\mathbf{I} \end{bmatrix}, \quad (3)$$

where V_A is the modulation variance of Alice, T is the overall transmittance between Alice and Bob, and χ_{line} is the noise in the channel line expressed in shot noise units. $\mathbf{I}$ and σ_z are the identity matrix, $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, and the Pauli matrix, $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, respectively.

The asymptotic limit SKR [bits/pulse] is calculated as

$$\text{SKR}_{asy} = \beta I_{AB} - S_{BE}, \quad (4)$$

where β is the reconciliation efficiency, I_{AB} is the mutual information between Alice and Bob, and S_{BE} upper bounds the Holevo information that represents the maximum mutual information between Eve and Bob in the protocol [15]. The mutual information for homodyne and heterodyne detection are calculated as

$$\begin{aligned} I_{AB, \text{hom}} &= \frac{1}{2} \log_2 \frac{(V_A + 1) + \chi_{\text{tot}}}{1 + \chi_{\text{tot}}}, \\ I_{AB, \text{het}} &= \log_2 \frac{(V_A + 1) + \chi_{\text{tot}}}{1 + \chi_{\text{tot}}}, \end{aligned} \quad (5)$$

where the total excess noise, χ_{tot} , combines both the channel noise, $\chi_{\text{line}} = \frac{1}{T} - 1 + \epsilon_{ch}$ (where ϵ_{ch} is the channel excess noise (Table I)), and detection excess noise, $\chi_{\text{hom/het}}$, and is expressed as $\chi_{\text{tot}} = \chi_{\text{line}} + \frac{\chi_{\text{hom/het}}}{T}$ [19]. The detection noise is different for homodyne and heterodyne detection where $\chi_{\text{hom}} = \frac{(1-\eta)+\epsilon_{det}}{\eta}$ and $\chi_{\text{het}} = \frac{1+(1-\eta)+2\epsilon_{det}}{\eta}$, respectively. Here, η is the detector efficiency and ϵ_{det} is the detector excess noise (Table I).

The Holevo information is calculated as

$$S_{BE} = G\left(\frac{\lambda_1 - 1}{2}\right) + G\left(\frac{\lambda_2 - 1}{2}\right) - G\left(\frac{\lambda_3 - 1}{2}\right) - G\left(\frac{\lambda_4 - 1}{2}\right), \quad (6)$$

where $G(x) = (x + 1) \log_2(x + 1) - x \log_2 x$ and λ are the symplectic eigenvalues of the covariance matrix, γ_{AB} . $\lambda_{1,2}$ is calculated as

$$\lambda_{1,2} = \sqrt{\frac{1}{2}(A \pm \sqrt{A^2 - 4B})}, \quad (7)$$

where

$$A = (V_A + 1)^2 + T^2(V_A + 1 + \chi_{\text{line}})^2 - 2TZ^2, \text{ and} \\ B = (T(V_A + 1)^2 + T(V_A + 1)\chi_{\text{line}} - TZ^2)^2. \quad (8)$$

$\lambda_{3,4}$ is calculated as

$$\lambda_{3,4} = \sqrt{\frac{1}{2}(C \pm \sqrt{C^2 - 4D})}, \quad (9)$$

where

$$C_{\text{hom}} = \frac{A\chi_{\text{hom}} + (V_A + 1)\sqrt{B} + T(V_A + 1 + \chi_{\text{line}})}{T(V_A + 1 + \chi_{\text{tot}})}, \\ D_{\text{hom}} = \sqrt{B} \frac{V_A + 1 + \sqrt{B}\chi_{\text{hom}}}{T(V_A + 1 + \chi_{\text{tot}})}, \quad (10)$$

for homodyne detection, and

$$C_{\text{het}} = \frac{A\chi_{\text{het}}^2 + B + 1 + 2TZ^2}{[T(V_A + 1 + \chi_{\text{tot}})]^2}, \\ + \frac{2\chi_{\text{het}}[(V_A + 1)\sqrt{B} + T(V_A + 1 + \chi_{\text{line}})]}{[T(V_A + 1 + \chi_{\text{tot}})]^2}, \\ D_{\text{het}} = \left(\frac{V_A + 1 + \sqrt{B}\chi_{\text{het}}}{T(V_A + 1 + \chi_{\text{tot}})} \right)^2 \quad (11)$$

for heterodyne detection. Here, $Z = \sqrt{V_A^2 + 2V_A}$ for Gaussian modulation.

B. M -PSK DM-CVQKD

In the 2-PSK protocol, Alice sends one of two coherent states with equal probability (0.5). In the 4-PSK protocol, Alice sends one of four coherent states with equal probability (0.25). In the 8-PSK protocol, Alice sends one of eight coherent states with equal probability (0.125). This is summarised below [18]:

- $\mathcal{S}_2 = \{|\alpha\rangle, |\alpha e^{i\pi}\rangle\}$
- $\mathcal{S}_4 = \{|\alpha\rangle, |\alpha e^{i\pi/2}\rangle, |\alpha e^{i\pi}\rangle, |\alpha e^{i3\pi/2}\rangle\}$
- $\mathcal{S}_8 = \{|\alpha\rangle, |\alpha e^{i\pi/4}\rangle, |\alpha e^{i\pi/2}\rangle, |\alpha e^{i3\pi/4}\rangle, |\alpha e^{i\pi}\rangle, |\alpha e^{i5\pi/4}\rangle, |\alpha e^{i3\pi/2}\rangle, |\alpha e^{i7\pi/8}\rangle\}$

The M -PSK ($M = 2, 4, 8$) protocols use coherent states with a magnitude α , and are represented in the optical phase space as shown in Figure 1.

The DM-CVQKD covariance matrix which describes the discretely modulated coherent state sent from Alice to Bob,

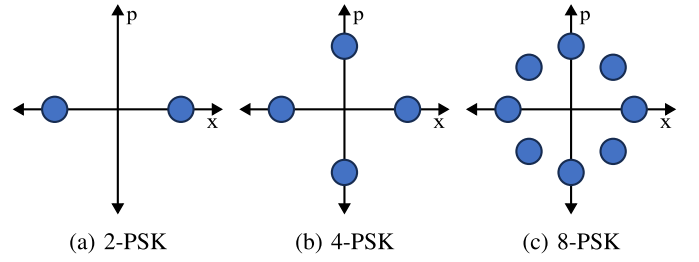


Fig. 1. Constellation diagrams of the 2,4,8-PSK protocols on the optical phase space. The coherent states have been modulated with a constant α and have equal probability. x = amplitude quadrature, p = phase quadrature.

with security under collective attacks, has the same form as the Gaussian modulation scheme [19],

$$\gamma_{AB} = \begin{bmatrix} (V_A + 1)\mathbf{I} & \sqrt{T}Z_M\boldsymbol{\sigma}_z \\ \sqrt{T}Z_M\boldsymbol{\sigma}_z & T(V_A + 1 + \chi_{\text{line}})\mathbf{I} \end{bmatrix}. \quad (12)$$

In this case, the correlation coefficient, Z_M , varies between each M -PSK protocol:

- $Z_2 = \alpha^2(\zeta_0^{3/2}\zeta_1^{-1/2} + \zeta_1^{3/2}\zeta_0^{-1/2})$,
- $Z_4 = 2\alpha^2 \sum_{k=0}^3 (\zeta_{k-1}^{3/2}\zeta_k^{-1/2})$,
- $Z_8 = 2\alpha^2 \sum_{k=0}^7 (\zeta_{k-1}^{3/2}\zeta_k^{-1/2})$.

The parameter ζ_k varies for each M -PSK protocol:

2-PSK protocol:

- $\zeta_0 = e^{-\alpha^2} \cosh \alpha^2$,
- $\zeta_1 = e^{-\alpha^2} \sinh \alpha^2$.

4-PSK protocol:

- $\zeta_{0,2} = \frac{1}{2}e^{-\alpha^2}(\cosh \alpha^2 \pm \cos \alpha^2)$,
- $\zeta_{1,3} = \frac{1}{2}e^{-\alpha^2}(\sinh \alpha^2 \pm \sin \alpha^2)$.

8-PSK protocol:

- $\zeta_{0,4} = \frac{1}{4}e^{-\alpha^2}(\cosh \alpha^2 + \cos \alpha^2 \pm 2 \cos \frac{\alpha^2}{\sqrt{2}} \cosh \frac{\alpha^2}{\sqrt{2}})$,
- $\zeta_{1,5} = \frac{1}{4}e^{-\alpha^2}(\sinh \alpha^2 + \sin \alpha^2 \pm \sqrt{2} \cos \frac{\alpha^2}{\sqrt{2}} \sinh \frac{\alpha^2}{\sqrt{2}} \pm \sqrt{2} \sin \frac{\alpha^2}{\sqrt{2}} \cosh \frac{\alpha^2}{\sqrt{2}})$,
- $\zeta_{2,6} = \frac{1}{4}e^{-\alpha^2}(\cosh \alpha^2 - \cos \alpha^2 \pm 2 \sin \frac{\alpha^2}{\sqrt{2}} \sinh \frac{\alpha^2}{\sqrt{2}})$,
- $\zeta_{3,7} = \frac{1}{4}e^{-\alpha^2}(\sinh \alpha^2 - \sin \alpha^2 \mp \sqrt{2} \cos \frac{\alpha^2}{\sqrt{2}} \sinh \frac{\alpha^2}{\sqrt{2}} \pm \sqrt{2} \sin \frac{\alpha^2}{\sqrt{2}} \cosh \frac{\alpha^2}{\sqrt{2}})$,

where $\alpha = \sqrt{\frac{V_A}{2}}$.

The asymptotic limit SKR is calculated as shown in Equations 4-11. However, Z_M is used for the Z term.

C. M -QAM

In [28], a security proof for the M -PSK and M -QAM DM-CVQKD protocols without Gaussian optimality under collective attacks from an eavesdropper in the asymptotic limit was developed. Unfortunately, using the security proof with the transmittances and excess noise in a typical satellite-to-ground link (Table I), M -PSK is not capable of producing a positive SKR. A more attractive protocol is M -QAM as it produces positive SKRs at lower transmittances and higher levels of excess noise. In M -QAM, M coherent states are modulated to be distributed equidistantly with each other on the optical phase space. By assigning a non-uniform probability to each coherent state, M -QAM can be tailored to a

discretised Gaussian distribution to further increase the mutual information between Alice and Bob.

The modulated coherent state is described as

$$\alpha_{k,l} = \frac{\alpha\sqrt{2}}{\sqrt{m-1}} \left(k - \frac{m-1}{2} \right) + i \frac{\alpha\sqrt{2}}{\sqrt{m-1}} \left(l - \frac{m-1}{2} \right), \quad (13)$$

where $M = m^2$ and the coherent states are equidistantly spaced between $-\sqrt{m-1}$ and $\sqrt{m-1}$ in the phase and amplitude quadratures. Here, $k = l = 0, 1, \dots, (m-1)$. The probability of each coherent state, $p_{k,l}$, can follow either a binomial distribution,

$$p_{k,l} = \frac{1}{2^{2(m-1)}} \binom{m-1}{k} \binom{m-1}{l}, \quad (14)$$

or a discrete Gaussian distribution,

$$p_{k,l} \sim \exp(-v(x^2 + p^2)), \quad (15)$$

where $x = \frac{\alpha\sqrt{2}}{\sqrt{m-1}} (k - \frac{m-1}{2})$ and $p = \frac{\alpha\sqrt{2}}{\sqrt{m-1}} (l - \frac{m-1}{2})$. Here, v is a free parameter which is optimised to maximise the SKR. For example, Figure 2 shows 16-QAM, 64-QAM, and 256-QAM on the optical phase space with the probability of each modulated coherent state based on the binomial distribution. It can be seen that a greater number of coherent states better approximates the Gaussian distribution.

The asymptotic limit SKR is calculated as in Equation 4. However, the mutual information is calculated as

$$I_{AB,\text{hom}} = \frac{1}{2} \log_2 \left(1 + \frac{TV_A}{2 + T\epsilon} \right),$$

$$I_{AB,\text{het}} = \log_2 \left(1 + \frac{TV_A}{2 + T\epsilon} \right). \quad (16)$$

The Holevo information is determined from the covariance matrix given by,

$$\Gamma_{AB}^* = \begin{bmatrix} (V_A + 1) \mathbf{I} & Z^* \sigma_z \\ Z^* \sigma_z & (1 + TV_A + T\epsilon) \mathbf{I} \end{bmatrix}. \quad (17)$$

The Holevo information is therefore,

$$S_{BE} = G \left(\frac{\lambda_1 - 1}{2} \right) + G \left(\frac{\lambda_2 - 1}{2} \right) - G \left(\frac{\lambda_3 - 1}{2} \right), \quad (18)$$

where λ_1 and λ_2 are the symplectic eigenvalues of the covariance matrix, Γ_{AB}^* . The symplectic eigenvalue, λ_3 , is calculated as

$$\lambda_{3,\text{hom}} = \sqrt{(V_A + 1) \left(V_A + 1 - \frac{Z^{*2}}{1 + TV_A + T\epsilon} \right)}, \text{ and}$$

$$\lambda_{3,\text{het}} = V_A + 1 - \frac{Z^{*2}}{2 + TV_A + T\epsilon}, \quad (19)$$

for homodyne and heterodyne detection, respectively. For an arbitrary modulation, the lower bound of the correlation coefficient, Z^* , can be calculated as

$$Z^*(T, \epsilon) = 2\sqrt{T} \text{Tr}(\tau^{\frac{1}{2}} \hat{a} \tau^{\frac{1}{2}} \hat{a}^\dagger) - \sqrt{2T\epsilon w}, \quad (20)$$

where $\hat{a}$ and $\hat{a}^\dagger$ are the annihilation and creation operators, respectively, ϵ is the total excess noise, and τ is the density matrix of the modulation:

$$\tau = \sum_k p_k |\alpha_k\rangle \langle \alpha_k|. \quad (21)$$

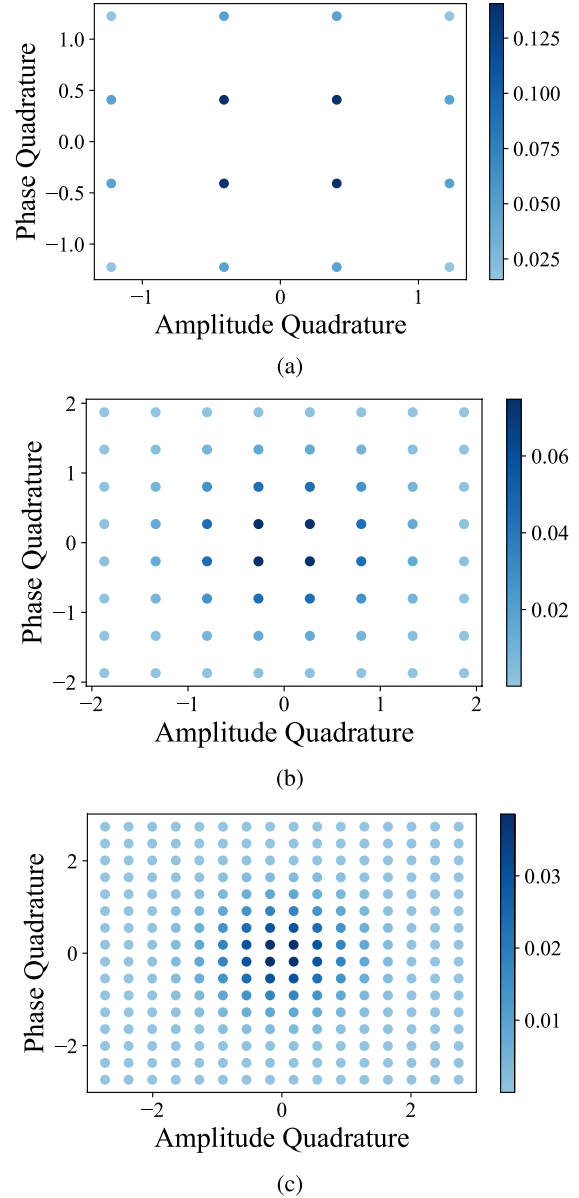


Fig. 2. M -QAM with probabilities based on the binomial distribution. (a) 16-QAM (b) 64-QAM (c) 256-QAM.

w is defined as

$$w = \sum_k p_k \left(|\alpha_k\rangle \hat{a}_\tau^\dagger \hat{a}_\tau |\alpha_k\rangle - |\alpha_k| \hat{a}_\tau |\alpha_k\rangle^2 \right). \quad (22)$$

Although M -QAM is a DM-CVQKD protocol, the assignment of a probability following a Gaussian distribution to each modulated coherent state raises the question of whether or not it loses its DM-CVQKD features, especially when M is large. In particular, the distinguishability of the coherent states on the optical phase space. In addition, the calculation of SKRs in the finite size limit for M -QAM remains an open question [28]. As a result, only the SKRs in the asymptotic limit will be studied in a satellite-to-ground context.

III. FINITE SIZE EFFECTS

The SKR calculation (Equation 4-11) is for the idealistic asymptotic limit in which an infinite number of symbols are sent between Alice and Bob, and provides an upper bound

to the achievable SKR. The finite size limit SKR, which represents a realistic case in which a finite number of symbols are sent between Alice and Bob [32], can be calculated as

$$\text{SKR}_{\text{fin}} = f(1 - \text{FER})(1 - v)[\beta I_{AB} - S_{BE} - \delta n_{\text{privacy}}], \quad (23)$$

where f is the laser repetition rate, FER is the frame error rate, v is the fraction of symbols excluded for channel parameter estimation, and $\delta n_{\text{privacy}}$ represents the proportion of the key further attributed to information gained by the eavesdropper to reflect the validity of estimated channel parameters in determining the Holevo information. However, to account for high error rate error correcting codes in long-distance CVQKD, the finite size limit SKR must be amended to [33], [34]:

$$\text{SKR}_{\text{fin}} = f[(1 - \text{FER})\beta I_{AB} - S_{BE} - \delta n_{\text{privacy}}]. \quad (24)$$

It should be noted that the term $(1 - v)$ has been excluded as shown in composable security proofs where Alice can estimate channel parameters after error correction [35], [36]. Here, $\delta n_{\text{privacy}}$ is calculated as

$$\delta n_{\text{privacy}} = \frac{(d+1)^2}{\sqrt{N}} + \frac{4(d+1)\sqrt{\log_2(\frac{2}{\epsilon_s})}}{\sqrt{N}} + \frac{2\log_2(\frac{2}{\epsilon^2\epsilon_s})}{\sqrt{N}} + \frac{\frac{4\epsilon_s d}{\epsilon\sqrt{N}}}{\sqrt{N}}, \quad (25)$$

where d is a discretisation parameter, ϵ_s is a smoothing parameter, ϵ is a security parameter representing the probability that the key is not secret, and N is the total number of symbols sent between Alice and Bob, the details of which can be found in previous work [31], [37], [38], [39]. Previous experimental CVQKD demonstrations have used laser repetition rates between 10 and 500 MHz [40], [41], [42], [43]. A laser repetition rate of $f = 50$ MHz has been chosen for the calculation of finite size limit SKRs, which is near the lower end of the range of laser repetition rates used in successfully demonstrated CVQKD, to show modest and more achievable finite size limit SKRs.

Reconciliation methods can be split into Multidimensional (MD) reconciliation and Multilevel Coding and Multistage Decoding (MLC-MSD) reconciliation [32]. MLC-MSD reconciliation is generally employed for CVQKD at SNRs greater than 0 dB, while MD reconciliation is employed for CVQKD at SNRs less than 0 dB. Low-density parity-check (LDPC) codes can be used with MD and MLC-MSD reconciliation. Multiedge-type LDPC (MET-LDPC) codes are regarded as suitable for both MD and MLC-MSD reconciliation. Recent practical MD reconciliation efficiencies and FERs asymptotically approach 100% as the SNR decreases [32].

The dependency of β and FER on the SNR in (Equation 26) have been empirically determined from [32], and are used to calculate and analyse the satellite-to-ground SKRs in the finite size limit. The FER equation is for $N = 10^6$. However, the general trend shows that the FER decreases as N increases [32]. As a result, a code block length of $N = 10^{11}$ has been used for finite size limit SKR calculations.

TABLE II
COEFFICIENTS OF β

Coefficient	MLC-MSD	MD
c_1	0.9655	-0.0825
c_2	0.0001507	0.1834
c_3	-0.04696	0.9821
c_4	-0.2238	-0.00002815

The coefficients, c_i , for β are displayed in Table II. The calculation of the FER uses the coefficients $m_1 = 0.8218$, $m_2 = -19.46$, and $m_3 = -298.1$ as shown in Equation 26.

$$\beta_{\text{MLC-MSD/MD}} = c_1^{c_2 \text{SNR}} - c_3^{c_4 \text{SNR}},$$

$$\text{FER} = \frac{1}{2}(1 + m_1 \arctan(m_2 \text{SNR} + m_3)). \quad (26)$$

The SNR is calculated as

$$\text{SNR} = 10 \log_{10} \left(\frac{T|\alpha|^2}{|\alpha|^2 + (1 - T)\chi_{\text{tot}}} \right), \quad (27)$$

and depends on α , transmittance (T) and excess noise (χ_{tot}) between Alice and Bob. Note that $\beta_{\text{MLC-MSD}}$, β_{MD} , and FER in Equation 26 are only valid when they have a value between 0 and 1.

IV. SATELLITE-TO-GROUND CHANNEL MODEL

Clouds, atmospheric turbulence, and atmospheric aerosols are the three main sources of signal degradation in the atmosphere that cause the transmittance of the signal to decrease through attenuation. The model developed considers the following: geometric losses due to the distance between Alice (satellite) and Bob (OGS), the hardware used, scintillation losses due to the atmospheric turbulence, scattering losses due to atmospheric aerosols. Clouds are not included in the model as they effectively destroy the signal and act as a blockage, completely attenuating the signal. The solution to this is accurate cloud coverage analyses and OGS network site diversity [44] to spatiotemporally maximise channel links between Alice and Bob regardless of environmental conditions. In addition, weather conditions such as rain, snow, and hail effectively destroy the signal and can cause damage to an OGS telescope [45]. Therefore, in the presence of clouds, rain, snow, and hail, satellite-to-ground CVQKD is not ideal.

Atmospheric losses depend on the thickness of the atmosphere or the atmospheric mass the signal propagates through. On average, 95% of the total atmosphere mass is within the first 20 km from ground to zenith [46], [47] and so atmospheric losses can be modelled as approximately confined to this range.

Figure 3 displays the channel link between the satellite and OGS where the total link distance and effective atmosphere thickness is a function of the elevation angle. The model assumes a uniform atmosphere thickness of 20 km. The total link distance, L_{tot} , and effective atmosphere thickness, $L_{\text{atm,eff}}$, can be calculated as

$$L_{\text{tot}} = ((R_E + L_{\text{zen}})^2 + (R_E + L_{\text{OGS}})^2 - 2(R_E + L_{\text{zen}})(R_E + L_{\text{OGS}})\cos(\alpha_1))^{\frac{1}{2}}$$

$$\alpha_1 = \arcsin \left[\cos(\theta) \frac{(R_E + L_{\text{OGS}})}{R_E + L_{\text{zen}}} \right] + (90 - \theta), \text{ and}$$

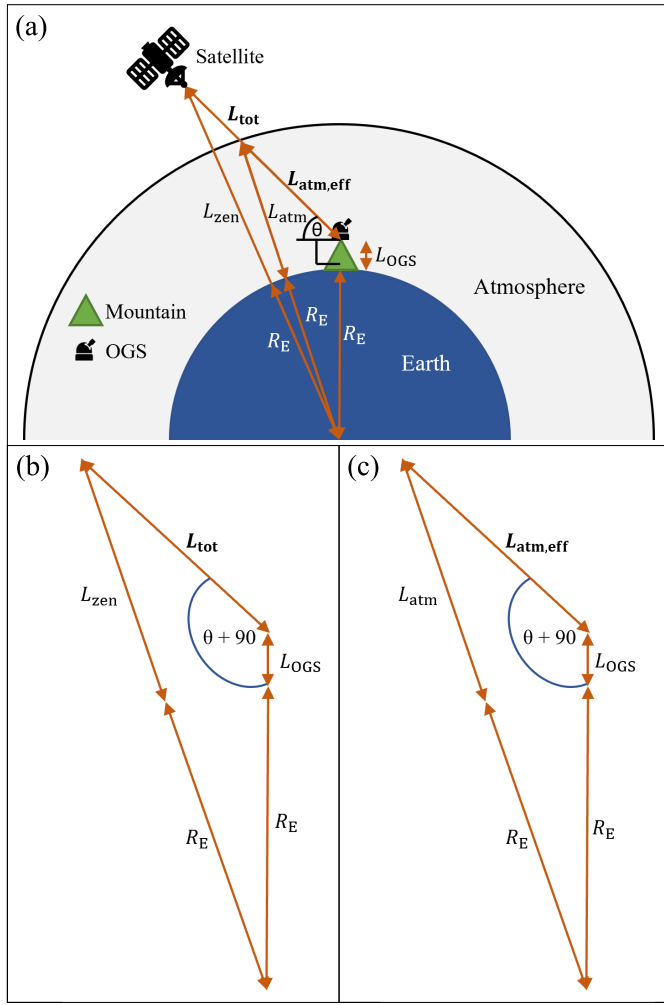


Fig. 3. The satellite-to-ground channel model is shown in (a). (b) and (c) shows the trigonometric determination of the total link distance and effective atmosphere thickness, respectively.

$$L_{\text{atm,eff}} = ((R_E + L_{\text{atm}})^2 + (R_E + L_{\text{OGS}})^2 - 2(R_E + L_{\text{atm}})(R_E + L_{\text{OGS}})\cos(\alpha_2))^{\frac{1}{2}}$$

$$\alpha_2 = \arcsin \left[\cos(\theta) \frac{(R_E + L_{\text{OGS}})}{R_E + L_{\text{atm}}} \right] + (90 - \theta), \quad (28)$$

where R_E is the radius of Earth, L_{OGS} is the altitude of the OGS, L_{zen} is the altitude of the satellite at zenith (90° elevation angle), L_{atm} is the atmospheric thickness containing 95% of atmospheric mass, and $\theta > 0$ is the elevation angle.

A. Geometric Losses

Geometric losses, A_{geo} (in units of decibels), arise from the link distance and optical hardware used in the transmission and reception of the signal, and can be estimated from [48],

$$A_{\text{geo}} = 10 \log_{10} \left(\frac{L_{\text{tot}}^2 \lambda^2}{D_t^2 D_r^2} \frac{1}{T_t(1 - L_p)T_r} \right) [\text{dB}], \quad (29)$$

which depends on the total link distance (L_{tot}), wavelength (λ), transmitter and receiver aperture diameter (D_t, D_r), transmitter and receiver efficiencies (T_t, T_r), and pointing loss (L_p) which is attributed to the inefficiency of the acquisition,

pointing, tracking (APT) system of the OGS as well as beam wandering. Equation 29 assumes that the receiver is in the far field of the transmitter ($L_{\text{tot}} \geq \frac{D_r D_t}{\lambda}$), the transmitter is diffraction limited, and there is no attenuation from the atmosphere which enables the later addition of other atmospheric losses (scattering and scintillation losses) in the overall model.

B. Scattering Losses

Atmospheric aerosols attenuate the signal through three regimes of scattering [49]: Rayleigh scattering caused by air molecules, Mie scattering caused by haze and fog, and geometrical scattering caused by rain, snow etc. For the 1550 nm wavelength used in CVQKD, the effects of Rayleigh scattering on the transmittance are negligible [49], and geometrical scattering effects are neglected as satellite-to-ground CVQKD, in general, is not ideal during weather conditions such as rain, snow, and hail.

The Kruse and Kim model [50] is used to model losses related to Mie scattering. The model depends on the wavelength of the signal and the atmospheric visibility, V (Equation 30). The resulting loss per kilometre is then multiplied by the effective atmosphere thickness (Equation 28):

$$A_{\text{scat}} = 10 \log_{10}(e) \left(\frac{3.912}{V} \right) \left(\frac{\lambda}{550} \right)^{-p} [\text{dB/km}],$$

$$p = \begin{cases} 1.6 & V \geq 50 \text{ km} \\ 1.3 & 6 \text{ km} \leq V < 50 \text{ km} \\ 0.16V + 0.34 & 1 \text{ km} \leq V < 6 \text{ km} \\ V - 0.5 & 0.5 \text{ km} \leq V < 1 \text{ km} \\ 0 & V < 0.5 \text{ km} \end{cases} \quad (30)$$

C. Scintillation Losses

Atmospheric turbulence causes refractive index variations in the atmosphere and distorts the optical wavefront of the laser, leading to intensity fluctuations and losses in the received signal observed as speckles on an imaging detector [44]. This is a phenomenon known as scintillation, and it can be quantified by the scintillation index, σ_I^2 . As the receiver involves a telescope, it is assumed to have an aperture diameter larger than the irradiance correlation width (lateral intensity coherence length) of the speckles. Therefore, aperture averaging is utilised, decreasing the adverse effects of scintillation [51].

The losses related to scintillation with aperture averaging are described in Equation 31 which depends on the scintillation index, σ_I^2 , and the probability that the received power is below the minimum required power to register a signal, p_{thr} . p_{thr} is equivalent to the fraction of link outage time. Here, it is assumed that the received signal is a spherical wave. The resulting loss as a function of the scintillation index is given by [51],

$$A_{\text{sci}} = 4.343 \left(\text{erf}^{-1}(2p_{\text{thr}} - 1) [2 \ln(\sigma_I^2 + 1)]^{\frac{1}{2}} - \frac{1}{2} \ln(\sigma_I^2 + 1) \right) [\text{dB}]. \quad (31)$$

The scintillation index can be calculated using Equation 32 which depends on the Rytov variance (σ_R^2), the effective

atmosphere thickness ($L_{\text{atm,eff}}$), and the refractive index structure parameter (C_n^2). Here, $k = \frac{2\pi}{\lambda}$ is the wave number, $d = D_r(\frac{\pi}{2\lambda L_{\text{atm,eff}}})^{\frac{1}{2}}$, D_r is the aperture diameter of the receiver, and λ is the wavelength.

$$\sigma_I^2(D_r) = \exp\left\{\frac{0.20\sigma_R^2}{[1 + 0.18d^2 + 0.20(\sigma_R^2)^{\frac{6}{5}}]^{\frac{7}{6}}} + \frac{0.21\sigma_R^2[1 + 0.24(\sigma_R^2)^{\frac{6}{5}}]^{-\frac{5}{6}}}{1 + 0.90d^2 + 0.21d^2(\sigma_R^2)^{\frac{6}{5}}}\right\} - 1, \\ \sigma_R^2 = 2.25k^{\frac{7}{6}} \int_0^{L_{\text{atm,eff}}} C_n^2(z)(L_{\text{atm,eff}} - z)^{\frac{5}{6}} dz. \quad (32)$$

The refractive index structure parameter, $C_n^2 [m^{-\frac{2}{3}}]$, defines the intensity of turbulence in the atmosphere. Although this can be calculated more accurately through models such as the Hufnagel-Valley models [5], [52], C_n^2 is set to be a constant parameter, where $C_n^2 = 10^{-16}$ corresponds to low turbulence levels, $C_n^2 = 10^{-14}$ corresponds to medium turbulence levels, and $C_n^2 = 10^{-13}$ corresponds to high turbulence levels [45].

V. SATELLITE-TO-GROUND GM-CVQKD AND DM-CVQKD SKR ANALYSIS

In this section, the SKRs of satellite-to-ground GM-CVQKD and DM-CVQKD are estimated with typical daylight excess noise values (Table I), and the losses applied to the total link distance and effective atmosphere thickness. The total attenuation based on losses is defined as

$$A_{\text{tot}} = A_{\text{geo}}(L_{\text{tot}}) + A_{\text{scat}}L_{\text{atm,eff}} + A_{\text{sci}}(L_{\text{atm,eff}}) \text{ [dB]} \quad (33)$$

and the transmittance is calculated as the non-logarithmic inverse of A_{tot} . Of the three sources of attenuation, geometric loss dominates due to the relatively large distance between the satellite and OGS. In calculating geometric losses, the assumption that the receiver is in the far field of the transmitter is governed by $L \geq \frac{D_r D_t}{\lambda}$. In situations where this inequality does not hold, the SKR calculation has been omitted. However, the general trend should still apply.

Several parameters were varied for the analysis of the GM-CVQKD and DM-CVQKD protocols under different satellite orbits (Table III). The two values for the visibility and the refractive index structure parameter correspond to good atmospheric conditions where the visibility is high and there is low atmospheric turbulence ($V = 200$ km, $C_n^2 = 10^{-16}$), and bad atmospheric conditions where visibility is low and there is high atmospheric turbulence ($V = 20$ km, $C_n^2 = 10^{-13}$). Two values of the receiver aperture diameter were also used to observe the changes in achievable SKR. The modulation variance for Gaussian and discrete modulation were chosen to be close to optimal ($V_A = 5$ SNU for Gaussian modulation, $V_A = 0.5$ SNU for M -PSK, $V_A = 2$ SNU for M -QAM). The SKRs for GM-CVQKD and DM-CVQKD were calculated as a function of the satellite altitude at zenith at different elevation angles (30° , 60° , 90°) in reference to the satellite altitude at zenith.

TABLE III
CVQKD LINK PARAMETERS

Parameter	Value (Unit)
Detection	Homodyne (M -PSK, GM-CVQKD) Heterodyne (M -QAM)
Modulation Variance (V_A)	0.5 SNU (M -PSK) 2 SNU (M -QAM) 5 SNU (Gaussian)
Laser repetition rate (f)	50 MHz
Discretisation parameter (d)	5
Smoothing parameter (ϵ_s)	2×10^{-10}
Security parameter (ϵ)	1×10^{-9}
Wavelength (λ)	1550 nm
Transmitter Aperture Diameter (D_t)	0.3 m
Receiver Aperture Diameter (D_r)	1, 2 m
Transmitter, Receiver Optics Efficiency (T_t, T_r)	0.9, 0.9
Pointing Loss Efficiency / APT Efficiency (L_p)	0.1
OGS Elevation (L_{OGS})	0 km 1.029 km (Mt. John)
Atmosphere Thickness (95% mass) (L_{atm})	20 km
Visibility (V)	20 km (Bad) 200 km (Good)
Refractive Index Structure Parameter (C_n^2)	$10^{-13} \text{ m}^{-\frac{2}{3}}$ (Bad) $10^{-16} \text{ m}^{-\frac{2}{3}}$ (Good)
Probability Threshold (p_{thr})	10^{-6}
Code block length	10^{11}
/ Total number of symbols sent (N)	

A. Asymptotic Limit SKRs of GM-CVQKD and DM-CVQKD Protocols

This section calculates the asymptotic limit SKRs for satellite-to-ground GM-CVQKD and DM-CVQKD under collective attacks from Eve. A reconciliation efficiency of 90% has been used for asymptotic limit SKR calculations. The asymptotic limit SKRs are also compared to the loss-limited SKR upper bound when the satellite is at zenith [29].

The results in Figure 4 show that for a receiver aperture diameter of 1 m and good atmospheric conditions, GM-CVQKD outperforms all DM-CVQKD protocols. The 2-PSK protocol does not produce any positive SKRs and is therefore unsuitable for satellite-to-ground downlinks. As expected, smaller elevation angles produce smaller SKRs as the signal travels through a larger link distance and effective atmosphere thickness, suffering greater attenuation. The M -QAM protocol outperforms the M -PSK protocol, where M -PSK can only achieve positive SKRs at smaller LEO altitudes. The M -QAM protocol can produce positive SKRs at small medium Earth orbits (MEOs - orbit altitudes of 1000-35,786 km) with the 256-QAM protocol capable of producing positive SKRs at larger link distances.

In bad atmospheric conditions, the M -PSK protocol cannot produce positive SKRs in LEO. In addition, the SKRs of GM-CVQKD and M -QAM have been significantly decreased (Figure 5).

Figure 4 and Figure 5 show that the asymptotic limit SKRs for GM-CVQKD and DM-CVQKD are well below the loss-limited upper bound of SKRs [29], [30]. This is because of the introduction of excess noise in the model used to calculate the SKRs as well as the other parameter values used in Table III, that would arise in practical satellite-to-ground CVQKD, which lower the achievable SKRs.

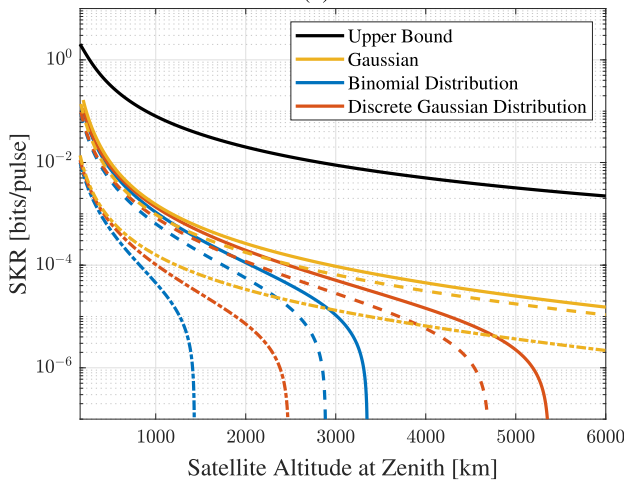
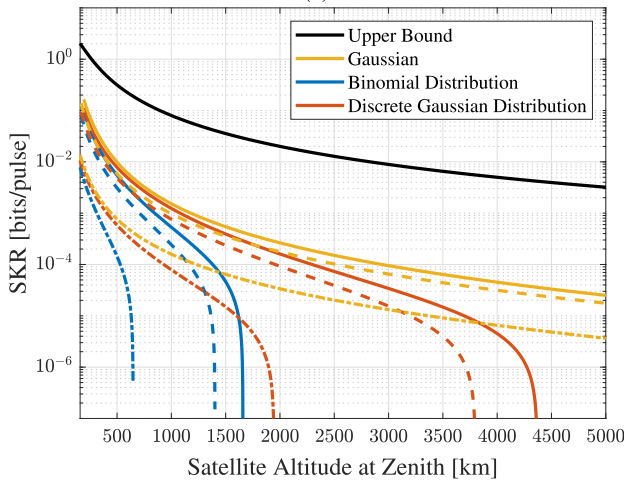
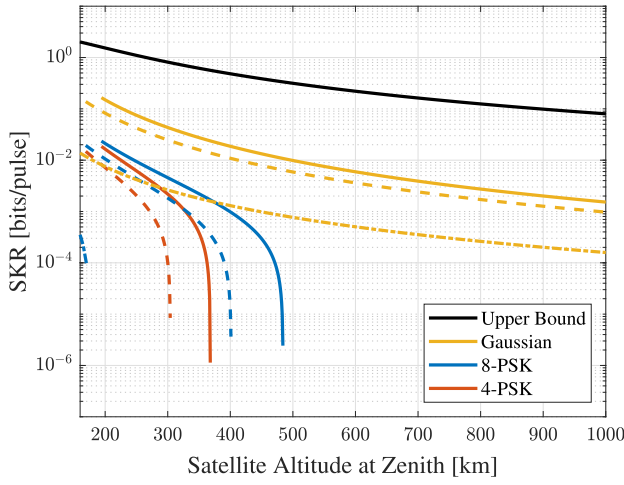


Fig. 4. Asymptotic limit SKRs as a function of satellite altitude for (a) M -PSK, (b) 64-QAM, and (c) 256-QAM DM-CVQKD in relation to GM-CVQKD in good atmospheric conditions. The solid lines indicate $\theta = 90^\circ$, dashed lines indicate $\theta = 60^\circ$, dash-dotted lines indicate $\theta = 30^\circ$. $D_r = 1$ m, $\beta = 90\%$.

B. Finite Size Limit SKRs of GM-CVQKD and DM-CVQKD

As security proofs for DM-CVQKD in the finite size limit are an active field of research [53], [54], [55], SKR analysis in the finite size limit was restricted to GM-CVQKD.

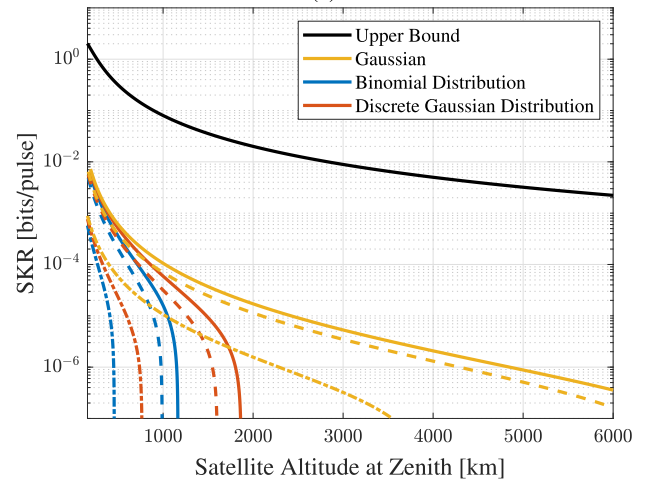
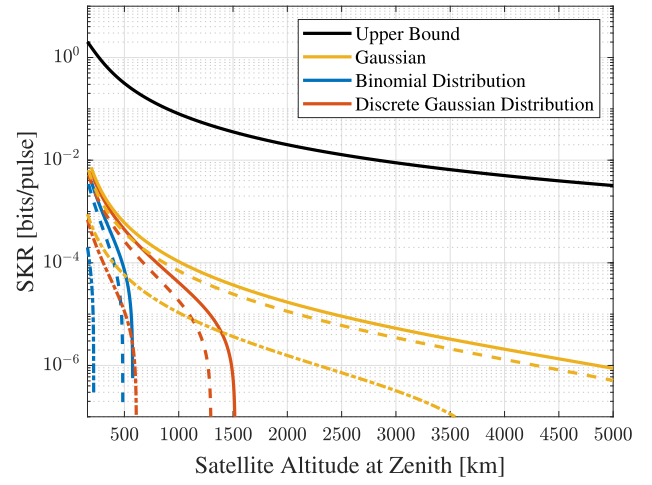


Fig. 5. Asymptotic limit SKRs as a function of satellite altitude for (a) 64-QAM and (b) 256-QAM DM-CVQKD in relation to GM-CVQKD in bad atmospheric conditions. The solid lines indicate $\theta = 90^\circ$, dashed lines indicate $\theta = 60^\circ$, dash-dotted lines indicate $\theta = 30^\circ$. $D_r = 1$ m, $\beta = 90\%$.

This analysis uses Equations 24-27. However, it is suspected that the relationship from asymptotic limit to finite size limit DM-CVQKD follows the same trend from asymptotic limit to finite size limit GM-CVQKD. No positive finite size limit SKRs are achieved in LEO GM-CVQKD in bad atmospheric conditions. This is evidence of the inability of CVQKD to operate in bad atmospheric conditions due to significant signal attenuation.

Figure 6 shows the finite SKRs in good atmospheric conditions for MD and MLC-MSD reconciliation. For a 1 m receiver aperture diameter, positive SKRs are restricted to a satellite orbit altitude below approximately 375 km and with larger elevation angles. Active propulsion is required to maintain this altitude, adding to the required complexities of a CVQKD LEO satellite. However, this situation can be ameliorated by increasing the receiver aperture diameter. In this case, the receiver aperture diameter was increased from 1 m to 2 m, increasing the satellite orbit altitude for which there is a positive SKR to a maximum of approximately 850 km and with lower elevation angles (Figure 6b). The results show that

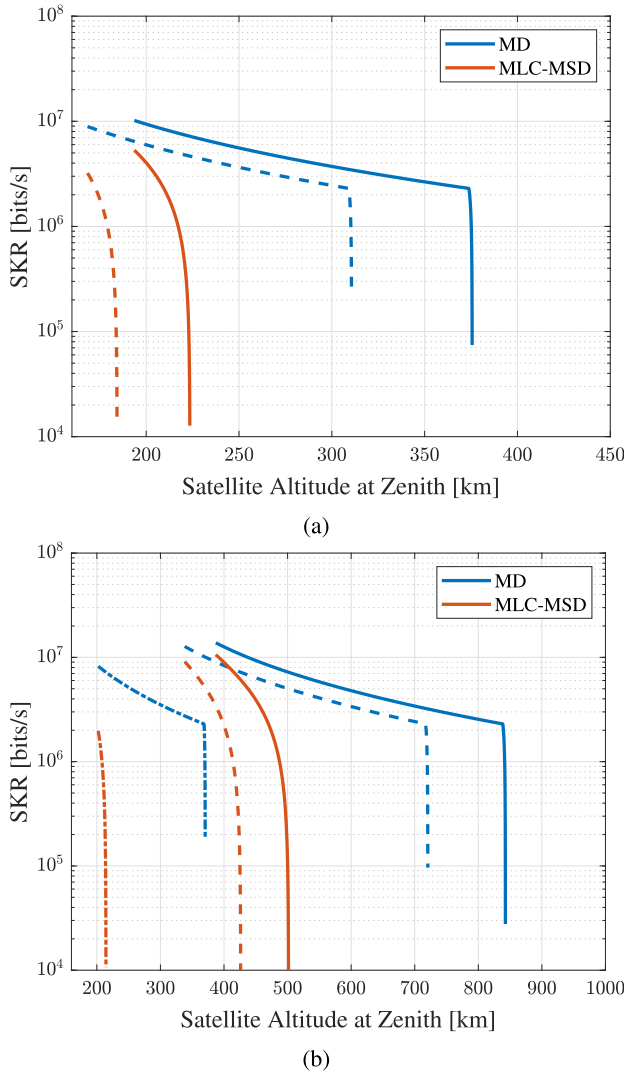


Fig. 6. Finite size limit SKRs as a function of satellite altitude for a (a) 1m and (b) 2 m receiver aperture diameter GM-CVQKD downlink using MD and MLC-MSD reconciliation with homodyne detection. The solid lines indicate $\theta = 90^\circ$, dashed lines indicate $\theta = 60^\circ$, dash-dotted lines indicate $\theta = 30^\circ$.

MD reconciliation outperforms MLC-MSD reconciliation by producing larger positive SKRs for larger link distances and lower elevation angles.

The finite size limit SKR calculation can also be presented as a function of elevation angle for a satellite at a certain orbit altitude. For this purpose, we assessed the orbit of the International Space Station (ISS), which has an average orbit altitude at zenith of 417.5 km, and studied a pass, shown in Figure 7, which occurred on 9th August 2022, over the University of Canterbury's Mt. John Observatory in New Zealand (Latitude = -43.9853° , Longitude = 170.4641° , Altitude = 1.029 km). The pass had a duration of 663 s with a maximum elevation angle of 87.6° . The calculations were made using the same link parameter values as in Table III with a receiver diameter aperture, D_r , of 2 m and OGS altitude, L_{geo} , of 1.029 km. A practical limitation to be noted is that a LEO satellite pass with large elevation angles may go through the keyhole of an observatory. In this situation, the satellite cannot be tracked.

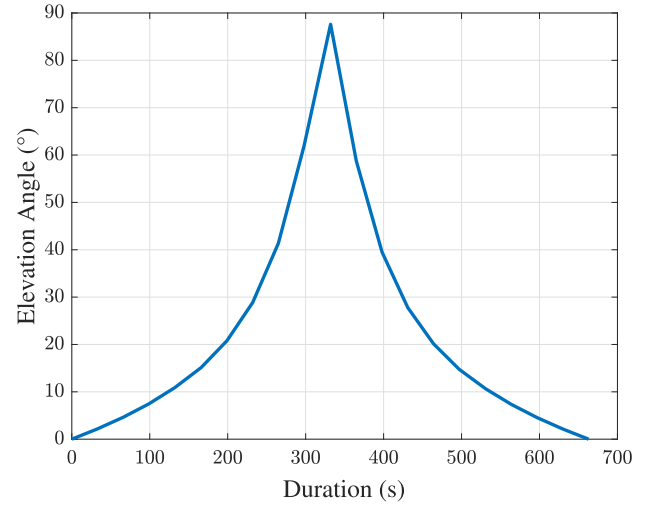


Fig. 7. Elevation for an ISS pass over Mt. John Observatory on 9th August 2022. The pass had a maximum elevation angle of 87.6° .

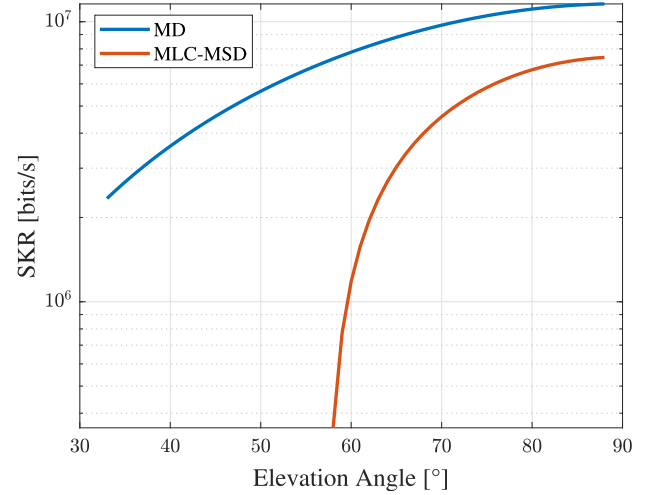


Fig. 8. SKR for ISS pass over Mt. John Observatory with maximum elevation angle 87.6° . $D_r = 2$ m. Homodyne detection. $L_{\text{geo}} = 1.029$ km.

The SKR was calculated assuming good atmospheric conditions for the ISS pass (Figure 8). The results show that finite size limit GM-CVQKD with MD reconciliation outperforms MLC-MSD reconciliation by producing larger positive SKRs and at lower elevation angles.

Multiplying the calculated SKR by the time that the ISS is at a certain elevation angle produces the total number of bits of the secret key. The elevation angle has been discretised with a resolution of 1° in Figure 7 to determine the temporal position of the ISS. The secret key has:

- 1.235 Gbit for GM-CVQKD with MD reconciliation,
- 385 Mbit for GM-CVQKD with MLC-MSD reconciliation.

C. SKR Trends From Parameter Variation Analysis

A sensitivity analysis of the parameters leads to the following observed trends:

- Increasing the laser repetition rate increases the finite size limit SKR.

- Decreasing the value for allowable information for Eve in the finite size limit ($\delta n_{\text{privacy}}$) increases the SKR for a given link distance. This implies that there is greater accuracy in the estimated amount of information Eve has intercepted (Holevo information).
- Increasing the transmitter and receiver aperture diameter as well as the transmitter and receiver optics efficiencies increases the maximum link distance and therefore satellite orbit altitude. This increase results from the decrease in losses during state preparation and measurement.
- Decreasing the pointing loss increases the maximum link distance and therefore satellite orbit altitude. This requires a more accurate APT system as well as less beam wandering between Alice and Bob.
- Increasing the OGS elevation increases the maximum link distance and therefore satellite orbit altitude for which a positive SKR is possible. This results from the decrease in attenuation from a decrease in link distance and effective atmosphere thickness as Bob is effectively closer to Alice.
- Operating in good atmospheric conditions where there is greater visibility and a smaller refractive index structure parameter increases the maximum link distance and therefore satellite orbit altitude. This results from the decrease in attenuation due to less atmospheric scattering and turbulence.
- Protocols with a larger number of coherent states lead to larger SKRs.

VI. DISCUSSION

The performance of recent MD and MLC-MSD reconciliation methods have been applied to GM-CVQKD in a satellite-to-ground context. It should be noted that the complexities of these reconciliation methods differ leading to different real-time decoding rates, and consequently, different times taken for real-time secret key generation between a satellite and OGS; the computational overhead of which would be more of a burden on the satellite. This stems from latency in data post-processing during the classical information reconciliation step in QKD. In combination with power consumption and heat dissipation in the satellite hardware, these practical elements would lead to a decrease in the achievable SKR. A practical solution to this would be to design an appropriate thermal management system on the satellite which would keep the satellite hardware to an appropriate low temperature to mitigate the negative effects of heating [56]. To solve power consumption and latency issues, the use of appropriate field-programmable gate arrays on the satellite for data post-processing would lead to relatively lower power consumption and high throughput by using their parallel computational ability [57]. Overall, a CVQKD satellite would need to be designed and engineered based on the desired SKRs during CVQKD. An alternative solution would be to perform information reconciliation offline and after the distribution of the secret key. Given that a LEO satellite pass is typically in the order of a few minutes, this is a promising solution for satellite-to-ground based QKD where the final secret key (after post-processing) is generated some time after the initial transmission, over multiple passes of the satellite over the OGS.

High turbulence in the atmosphere causes beam wandering during transmission between a satellite and an OGS. This may result in a large beam angle-of-arrival on the receiver aperture of the OGS, causing non-ideal coupling to fibre as the beam may not be focused on the fibre, and therefore this can introduce a probabilistic interruption in communication between the satellite and OGS. The details of this phenomena have previously been published [25], and it is similarly related to the probability threshold, p_{thr} , in Equation 31 which is the probability that the received power is below the minimum required power to register a signal, therefore causing an interruption in communication and a substantial decrease in achievable SKRs.

The loss-limited SKR upper bound was compared with achievable asymptotic limit SKRs for GM-CVQKD and DM-CVQKD. It should be noted that a thermal-loss SKR upper bound, which incorporates noise components, and a thermal-loss SKR lower bound have also been determined in References [29] and [30]. However, as shown in Figure 3 in Ref. [29], the loss-limited SKR upper bound and the thermal-loss SKR upper and lower bounds are identical in LEO and low MEO altitudes. For this reason, the loss-limited SKR upper bound was used for comparison. Furthermore, the bounds are for ideal systems. In contrast, the SKRs calculated for GM-CVQKD and DM-CVQKD consider parameters found in practical satellite-to-ground CVQKD such as the reconciliation efficiency as well as the modulation variance, which lead to lower but more practical achievable SKRs.

VII. CONCLUSION

In this work, a feasibility study for practical satellite-to-ground CVQKD with a focus on comparing existing GM-CVQKD and DM-CVQKD protocols with the application of the performance of recent reconciliation methods is presented. The results show that GM-CVQKD outperforms both the M -PSK and M -QAM DM-CVQKD protocols in the asymptotic limit. However, the M -QAM protocol outperforms the M -PSK protocol and is capable of producing positive SKRs in both LEO and low MEO. In the finite size limit, MD reconciliation outperforms MLC-MSD reconciliation by producing larger positive SKRs in GM-CVQKD for larger link distances and lower elevation angles. The physical implementations of both GM-CVQKD and DM-CVQKD as well as the processes for MD and MLC-MSD reconciliation have not been considered. These should be investigated further for future satellite-to-ground and long distance CVQKD operation.

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